1. **Train-Test Split**

- Use `train_test_split()` from `sklearn.model_selection` to divide your data.

- Training set: Used to train the model.

- Test set: Used to evaluate performance on unseen data.

2. **Generalization Error**

- Measures how well your model performs on new, unseen data.

- Low generalization error = good predictive power.

3. **Cross-Validation**

- Split the data into "folds". Train on some folds, test on the remaining.

- Repeat this for all folds.

- Average results for an estimate of out-of-sample error.

4. **Polynomial Order Selection**

- Low-order polynomial -> underfitting (too simple).

- High-order polynomial -> overfitting (too complex).

- Use test error to select best polynomial order.

5. **Test Error vs Polynomial Order**

- Plot mean square error vs polynomial order.

- Choose the order with the lowest test error.

6. **Ridge Regression**

- Use when independent variables are highly correlated.

- Prevents overfitting by penalizing large coefficients.

7. **Alpha in Ridge Regression**

- Alpha is a hyperparameter controlling coefficient magnitude.

- Split data into training and validation sets.

- Try different alpha values, compute R-squared on validation set.

- Choose alpha that gives highest R-squared.

8. **Grid Search & GridSearchCV()**

- Scans multiple hyperparameter values with cross-validation.

- Use `GridSearchCV()` from `sklearn.model_selection`.

- Pass a dictionary of hyperparameter names and values to try.